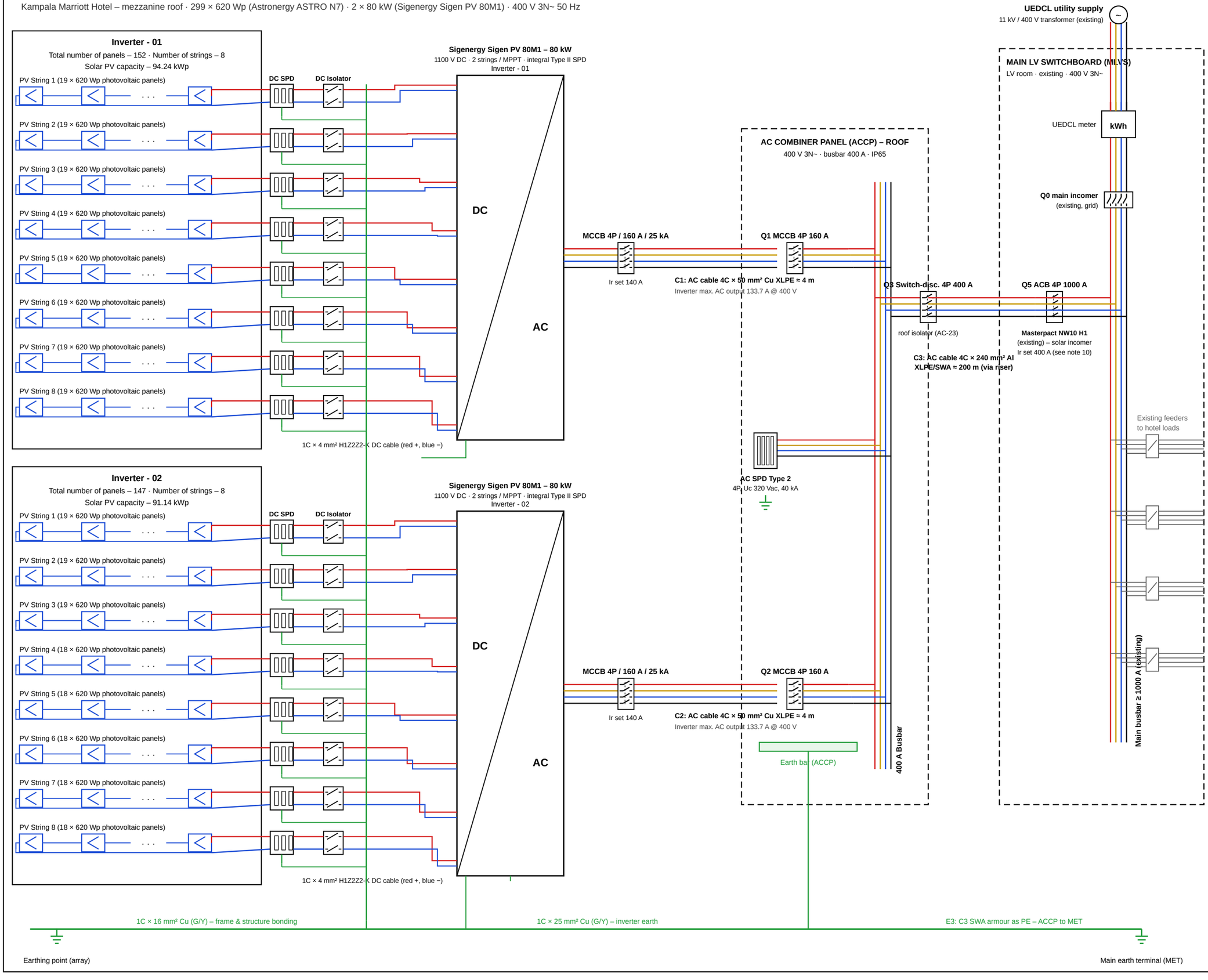
SINGLE LINE DIAGRAM – 185.4 kWp GRID-TIED ROOFTOP SOLAR PV SYSTEM

Kampala Marriott Hotel – mezzanine roof · 299 × 620 Wp (Astronergy ASTRO N7) · 2 × 80 kW (Sigenergy Sigen PV 80M1) · 400 V 3N~ 50 Hz



General Notes

This diagram shows the electrical schematic of the 185.4 kWp grid-tied solar PV system (no battery).

1. PV modules
Astronergy ASTRO N7 CHSM66RN(DG)/F-BH, 620 Wp bifacial; Voc 49.04 V, Isc 16.11 A, Vmp 41.56 V, Imp 14.92 A

2. Strings
INV-01: 8 × 19 = 152 (94.24 kWp); INV-02: 3 × 19 + 5 × 18 = 147 (91.14 kWp). Max string Voc ≈ 955 V (15 °C) < 1100 V

3. Inverters
2 × Sigenergy Sigen PV 80M1, 80 kW, 400 V 3N~; max AC current 133.7 A; 2 strings per MPPT (max 40 A/MPPT)

4. DC cables
1C × 4 mm² Cu H1Z2Z2-K, 1.5 kV DC, IEC 62930 / EN 50618, UV resistant

5. AC cables
C1, C2: 4C × 50 mm² Cu XLPE; C3: 4C × 240 mm² Al XLPE/SWA ≈ 200 m (spare stock), 0.6/1 kV, IEC 60502-1; VD ≈ 3.1 % at 232 A

6. Earthing cables
16 / 25 mm² Cu G/Y; ACCP to MET via C3 steel-wire armour, glands earthed at both ends (IEC 60364-5-54 adiabatic check)

7. DC SPD (per string)
Type 2, Ucpv 1100 V DC, In 20 kA, Imax 40 kA, Up ≤ 4.0 kV, IEC 61643-31

8. DC isolator (per string)
2-pole, 1200 V DC, 32 A, DC-PV2, IEC 60947-3, IP66 enclosure

9. AC SPD (ACCP)
Type 2, 4P (3P+N), Uc 320 Vac, In 20 kA, Imax 40 kA, Up ≤ 1.5 kV, IEC 61643-11

10. AC protection
Inverter MCCB 4P 160 A (Ir 140 A); ACCP Q1/Q2 MCCB 4P 160 A (Ir 140 A); Q3 switch-disconnector 4P 400 A. Q5 = existing Masterpact NW10 H1 ACB used as solar incomer, 1000 A sensor set at Ir 400 A (minimum); C3 4C × 240 mm² is rated above 400 A installed.

11. Solar system
PV capacity 185.38 kWp DC; inverter capacity 2 × 80 kW = 160 kW AC; DC/AC 1.16

12.
Cable lengths indicative; confirm on site. Verify MLVS spare way, fault level and busbar rating before installation.

Legend

- DC + / L1
- DC − / L3
- L2
- N
- Earth (PE)

Firm name & address	NileSun Power Ltd, Kampala, Uganda
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